

Problem 13.6.1

Show that

- a) $\sinh z = \sinh x \cos y + i \cosh x \sin y$
- b) $\cosh z = \cosh x \cos y + i \sinh x \sin y$

Solution

Problem 13.6.3

Show that

- a) $\cosh^2 z - \sinh^2 z = 1$
- b) $\cosh^2 z + \sinh^2 z = \cosh 2z$

Solution

Problem 13.6.7

Find, in the form $u + iv$

- a) $\cos i$
- b) $\sin i$

Solution

Problem 13.6.13

Using the definitions, prove

- a) $\cos z$ is even, $\cos(-z) = \cos z$
- b) $\sin z$ is odd, $\sin(-z) = -\sin z$

Solution

Problem 13.6.16

Find all solutions

$$\sin z = 100$$

Solution

Problem 13.7.5

Find

$$\operatorname{Ln}(-11)$$

Solution

Problem 13.7.7

Find

$$\operatorname{Ln}(4 - 4i)$$

Solution

Problem 13.7.8

Find

$$\operatorname{Ln}(1 \pm i)$$

Solution

Problem 13.7.15

Find all values and graph some in the complex plane

$$\ln(e^i)$$

Solution

Problem 13.7.19

Solve for z

$$\ln z = 4 - 3i$$

Solution

Problem 13.7.23

Find the principal value.

$$(1 + i)^{1-i}$$

Solution

Problem 14.1.1

Find and sketch the path

$$z(t) = \left(1 + \frac{1}{2}i\right)t \quad (2 \leq t \leq 5)$$

Solution

Problem 14.1.11

Find a parametric representation and sketch the path

Segment from $(-1, 1)$ to $(1, 3)$

Solution

Problem 14.1.19

Find a parametric representation and sketch the path

Parabola $y = 1 - \frac{1}{4}x^2$ $(-2 \leq x \leq 2)$

Solution

Problem 14.1.21

Integrate by the first method or state why it does not apply and use the second method. Show the details.

$$\int_C \operatorname{Re} z \, dz, \quad C \text{ the shortest path from } 1 + i \text{ to } 3 + 3i$$

Solution

Problem 14.1.25

$$\int_C z \exp(z^2) \, dz, \quad C \text{ from } 1 \text{ along the axes to } i$$

Solution

Problem 14.1.29

$$\int_C \operatorname{Im} z^2 \, dz \quad \text{counterclockwise around the triangle with vertices } 0, 1, i$$

Solution